Question: You are tasked with creating a database for a college system. The system will store information about students, courses, and grades. The registrar can generate reports of students who have not met the course requirements, and the dean has the authority to update the student's status. Perform the following tasks:

A. Write an SQL statement to create the database with the following tables:
   I. Student: Contains information about students, including a unique student ID (primary key), name, and email.
   II. Course: Stores information about courses, including a course ID (primary key), course title, and course description.
   III. Enrollment: Tracks student enrollments, including an enrollment ID (primary key), student ID, and course ID.
   IV. Grade: Stores student grades, including a grade ID (primary key), enrollment ID, and grade value.
B. Create two users, Registrar and Dean, and grant all rights on all tables to the Registrar user.
C. Revoke the delete right from the Registrar user on the Enrollment table.
D. Grant only the 'select' right to the Dean user.

Note: The field sizes mentioned are for reference and may vary based on the database system being used.

Solution

```
mysql> CREATE DATABASE CollegeSystem;
Query OK, 1 row affected (0.04 sec)

mysql> Show databases;
+--------------------+
| Database           |
+--------------------+
| CollegeSystem      |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)

mysql>
```

a) SQL statement to create the database with the tables:

Create the Student table:

CREATE TABLE Student (

StudentID NUMBER(10) PRIMARY KEY,

Name VARCHAR(100),

Email VARCHAR(100)

);

```
mysql> use CollegeSystem;
Database changed
mysql> CREATE TABLE Student (
    ->     StudentID INT PRIMARY KEY,
    ->     Name VARCHAR(255),
    ->     Email VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.32 sec)

mysql> show tables;
+-------------------------+
| Tables_in_CollegeSystem |
+-------------------------+
| Student                 |
+-------------------------+
1 row in set (0.02 sec)
```

Create the Course table:

CREATE TABLE Course (

CourseID VARCHAR(15) PRIMARY KEY,

CourseTitle VARCHAR(50),

CourseDescription VARCHAR(150)

);

```
mysql> CREATE TABLE Course (
    ->     CourseID INT PRIMARY KEY,
    ->     CourseTitle VARCHAR(255),
    ->     CourseDescription VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.14 sec)

mysql> show tables;
+-------------------------+
| Tables_in_CollegeSystem |
+-------------------------+
| Course                  |
| Student                 |
+-------------------------+
2 rows in set (0.01 sec)
```

Create the Enrollment table:

```
CREATE TABLE Enrollment (
   EnrollmentID NUMBER(10) PRIMARY KEY,
   StudentID NUMBER(10),
   CourseID VARCHAR(15),
   FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
   FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);
```

```
mysql> CREATE TABLE Enrollment (
    ->     EnrollmentID INT PRIMARY KEY,
    ->     StudentID INT,
    ->     CourseID INT,
    ->     FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    ->     FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
    -> );
Query OK, 0 rows affected (0.22 sec)

mysql> show tables;
+-------------------------+
| Tables_in_CollegeSystem |
+-------------------------+
| Course                  |
| Enrollment              |
| Student                 |
+-------------------------+
3 rows in set (0.00 sec)
```

Create the Grade table:

```
CREATE TABLE Grade (
   GradeID NUMBER(10) PRIMARY KEY,
   EnrollmentID NUMBER(10),
   GradeValue VARCHAR(2),
   FOREIGN KEY (EnrollmentID) REFERENCES Enrollment(EnrollmentID)
);
```

```
mysql> CREATE TABLE Grade (
    ->     GradeID INT PRIMARY KEY,
    ->     EnrollmentID INT,
    ->     GradeValue VARCHAR(5),
    ->     FOREIGN KEY (EnrollmentID) REFERENCES Enrollment(EnrollmentID)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> show tables;
+-------------------------+
| Tables_in_CollegeSystem |
+-------------------------+
| Course                  |
| Enrollment              |
| Grade                   |
| Student                 |
+-------------------------+
4 rows in set (0.00 sec)
```

b) Creating users and granting all rights to the Registrar user:

Create the Registrar user:

CREATE USER Registrar IDENTIFIED BY <password>;

Grant all rights on all tables to the Registrar user:

GRANT ALL PRIVILEGES ON Student, Course, Enrollment, Grade TO Registrar;

c) Revoking the delete right from the Registrar user on the Enrollment table:

Revoke delete right from the Registrar user on the Enrollment table:

REVOKE DELETE ON Enrollment FROM Registrar;

d) Granting only the 'select' right to the Dean user:

Create the Dean user:

CREATE USER Dean IDENTIFIED BY <password>;

Grant only the 'select' right to the Dean user on all tables:

GRANT SELECT ON Student, Course, Enrollment, Grade TO Dean;

```
mysql> -- Create the Registrar user
mysql> CREATE USER 'Registrar'@'localhost' IDENTIFIED BY 'Pass';
Query OK, 0 rows affected (0.07 sec)

mysql> -- Grant all rights on all tables to the Registrar user
mysql> GRANT ALL PRIVILEGES ON CollegeSystem.* TO 'Registrar'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> -- Create the Dean user
mysql> CREATE USER 'Dean'@'localhost' IDENTIFIED BY 'DPass';
Query OK, 0 rows affected (0.02 sec)

mysql> -- Grant all rights on all tables to the Dean user
mysql> GRANT SELECT ON CollegeSystem.* TO 'Dean'@'localhost';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> -- Grant only the 'select' right to the Dean user
mysql> GRANT SELECT ON CollegeSystem.* TO 'Dean'@'localhost';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> REVOKE DELETE ON CollegeSystem.Enrollment FROM 'Registrar'@'localhost';
Query OK, 0 rows affected (0.04 sec)
```

```
annas@annas-VirtualBox:~/Desktop$ mysql -u Dean -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.35-0ubuntu0.23.10.1 (Ubuntu)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use CollegeSystem
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> DELETE FROM Enrollment WHERE StudentID = 1 AND CourseID = 101;
ERROR 1142 (42000): DELETE command denied to user 'Dean'@'localhost' for table 'Enrollment'
mysql>
```

Note: **Please note that <password> should be replaced with the actual password for the respective users.**